The Laplace transform

Like many of the methods we've learned previously, the Laplace transform is just another method we can use to find the solution to a differential equation. Therefore, as always, our first priority is to understand when it's appropriate to use a Laplace transform, instead of one of the other techniques we already know.

For first and second order equations that are homogeneous (the forcing function on the right side of the equation is 0), it'll usually be easier to use the solution techniques we learned earlier. And even when the equation is nonhomogeneous, if the forcing function is simple enough, it probably won't be any faster or easier to apply a Laplace transform.

But Laplace transforms start to become more effective than other solution methods when the forcing function gets more complicated, and especially when the forcing function is discontinuous.

The definition of the Laplace transform

Think about a transform in general as some kind of operation that can be applied to a function that changes it in some kind of predictable way.

The derivative and the integral are transforms from calculus. Differentiating gives us a predictable function that models slope, while integrating gives us a predictable function that models area.

The Laplace transform is a specific integral transform. If we want to indicate the transform of a function $f(t)$, we write the Laplace transform as

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$\mathscr{L}(f(t))$. Taking the transform of $f(t)$ results in a predictable new function $F(s)$.

$$\mathscr{L}(f(t)) = F(s)$$

Keep in mind that we won't always use f to name the function we're transforming. For instance, it's also common to use y, in which case we might write the transform equation as

$$\mathscr{L}(y(x)) = Y(s)$$

In either case, the original function is defined in terms of some independent variable like t, x, etc.. When we apply the transform, that original variable will drop away, and the transform will be defined in terms of a new variable, s. And the transform function is named with a capital letter, such that if the original function was g, its transform is named G.

To calculate a function's transform, we can use the definition of the Laplace transform, which is

$$\mathscr{L}(f(t)) = F(s) = \int_0^\infty e^{-st} f(t)\, dt$$

This tells us that, given a function $f(t)$, we can find its transform $F(s)$ by multiplying $f(t)$ by e^{-st}, and then integrating that product on the interval $[0,\infty)$.

Let's do an example where we apply the definition in order to find the transform of a function.

Example

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Use the definition to find the Laplace transform of the function.

$$f(t) = e^{-2t}$$

Plugging the given function into the definition of the Laplace transform gives

$$F(s) = \int_0^\infty e^{-st} f(t)\ dt$$

$$F(s) = \int_0^\infty e^{-st} e^{-2t}\ dt$$

We can use the power rule for exponents to combine the exponentials.

$$F(s) = \int_0^\infty e^{-st-2t}\ dt$$

$$F(s) = \int_0^\infty e^{(-s-2)t}\ dt$$

Since we're integrating only with respect to t, we can treat s as a constant, which means the integral will be

$$F(s) = \frac{1}{-s-2} e^{(-s-2)t} \Big|_0^\infty$$

$$F(s) = \lim_{b \to \infty} \frac{1}{-s-2} e^{(-s-2)t} \Big|_0^b$$

Evaluate over the interval.

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$$F(s) = \lim_{b\to\infty}\left(\frac{1}{-s-2}e^{(-s-2)b}\right) - \frac{1}{-s-2}e^{(-s-2)0}$$

$$F(s) = \lim_{b\to\infty}\left(\frac{1}{-s-2}e^{(-s-2)b}\right) + \frac{1}{s+2}$$

From here, the value we find for the exponential will depend on the value of $-s-2$.

- If $-s-2>0$, the value of the exponential goes to ∞, and $F(s)$ diverges
- If $-s-2<0$, the value of the exponential goes to 0, and

$$F(s) = \frac{1}{s+2}$$

Because we only get a convergent value when $-s-2<0$, we'll assume

$$-s-2<0$$

$$-s<2$$

$$s>-2$$

in order to conclude that the Laplace transform of $f(t)=e^{-2t}$ is

$$F(s) = \frac{1}{s+2} \qquad \text{given } s>-2$$

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At the end of this last example, we assumed $s > -2$ in order to find the Laplace transform of $f(t) = e^{-2t}$. It's important to remember that all Laplace transforms will have these kinds of restrictions on s.

Because these restrictions always apply, they become part of our common understanding of Laplace transforms. Therefore, we won't always take the time to point out the restriction, and instead we'll just assume that we're all agreeing to use whatever restriction on s is required to make the transform function converge to a finite value.

Still, we'll look at this restriction more later on when we cover functions of exponential type.

An alternate definition of the transform

Sometimes we'll see a slightly different definition of the Laplace transform, written as

$$\mathscr{L}(f(t)) = F(s) = \int_{-\infty}^{\infty} e^{-st} f(t)\, dt$$

The only change to the original definition is a substitution in the lower limit. Instead of integrating on $[0,\infty)$, this new definition integrates on $(-\infty, \infty)$. If we use this definition with the $(-\infty, \infty)$ interval of integration, we're almost always assuming that the definition of the function $f(t)$ is given by

$$f(t) = \begin{cases} f(t) & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases}$$

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Because we're assuming $f(t) = 0$ whenever $t < 0$, the value of the integral on $(-\infty, 0)$ is 0. And by dropping that entire half of the integral to 0, we essentially just get back to the same definition of the transform that we started with.

We'll usually use a Heaviside function to force the behavior that $f(t) = 0$ whenever $t < 0$, but we'll talk more about that later. For now, it's just important that we know that these two definitions of the Laplace transform are essentially identical (because of the assumed value of $f(t)$), even though they appear to be slightly different.

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